



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 3 • STANDARD 6.AT.3

Ratio Reasoning: Convert Measurements Between Systems

Lesson 3-7 • Teacher Copy

Name: Type your name

Date: Today's date

Class: Class period



TODAY'S OBJECTIVES

Content Objective: I can use ratio reasoning to convert measurements between the customary and metric systems.

Language Objective: I can explain a conversion between systems using the words conversion factor, approximately, customary, and metric.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Convert Measurements Between Systems

Conversion factor

Customary system

Metric system

Approximately

Ratio table



Tap any word to see what it means and a picture.

1

Rewriting a customary measurement as a metric one gives an

answer, not an exact one.

2

The ratio that trades one unit for another is a

3

Inches, pounds and gallons belong to the

system.

4

Centimeters, grams and liters belong to the

system, built on tens.

5

Close to but not exactly equal is

— written with $\approx$.

6 A table whose every column holds the same comparison is a

 .

Write About the Math

Is 100 kilometers per hour faster or slower than 100 miles per hour, and how do you know?

¿100 kilómetros por hora es más rápido o más lento que 100 millas por hora, y cómo lo sabes?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Convert Measurements Between Systems**
Convertir medidas entre sistemas

☐ **Conversion factor**
Factor de conversión

☐ **Customary system**
Sistema usual

☐ **Metric system**
Sistema métrico

☐ **Approximately**
Aproximadamente

Model: A mile is longer than a kilometer ($1 \text{ km} \approx 0.6 \text{ mi}$), so 5 miles is the longer race even though the numbers match. — Students reason about UNIT SIZE before touching arithmetic — same number, different units, different lengths.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The conversion factor is $1 \text{ km} \approx \underline{\hspace{1cm}} \text{ mi}$. Scaling that up, 100 kilometers $\approx \underline{\hspace{1cm}}$ miles, so the sign means about $\underline{\hspace{1cm}}$ miles per hour. That is $\underline{\hspace{1cm}}$ than 100 miles per hour, because in one hour you would travel $\underline{\hspace{1cm}}$ distance.
 - My answer is $\underline{\hspace{1cm}}$ because $\underline{\hspace{1cm}}$.
Mi respuesta es $\underline{\hspace{1cm}}$ porque $\underline{\hspace{1cm}}$.
-
-

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is $\underline{\hspace{1cm}}$.

Mi afirmación es ____.

- **My evidence is** ____, **so** ____.

Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Convert Measurements Between Systems
2. **Notes 2:** Conversion factor
3. **Notes 3:** Customary system
4. **Notes 4:** Metric system
5. **Notes 5:** Approximately
6. **Notes 6:** Ratio table
7. **Watch for this mistake:** *A common mistake converting between systems is comparing the bare numbers without converting at all — deciding that 5 kilometers and 5 miles are the same distance because both say 5. They are not: 5 kilometers $\approx$ 3 miles. A second mistake is writing = instead of $\approx$. Conversion factors between systems are rounded (1 mile is about 1.609344 kilometers), so the result is approximate no matter how carefully you multiply.*
8. **Try It 1:** — *Each pair is a conversion factor between the two systems, and each one is approximate.*
9. **Try It 2:** A. The conversion factors are rounded, so results are approximate — *One mile is about 1.609344 kilometers — a decimal that never ends neatly. Any usable factor is rounded, so the answer is approximate.*